

Levi Newediuk

ASSISTANT PROFESSOR

Department of Biology, Mount Royal University

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Academic positions

Mount Royal University

ASSISTANT PROFESSOR

2025-Present

University of Manitoba

NSERC POSTDOCTORAL FELLOW

2025

University of Manitoba

POSTDOCTORAL FELLOW

2022-2024

Education

Memorial University of Newfoundland

PHD, BIOLOGY

Dr. Eric Vander Wal

2022

University of Manitoba

MSC, BIOLOGY

Dr. James Hare

2017

University of Manitoba

BSC HONS, BIOLOGY

Dr. James Hare

2011

Academic profile

RESEARCH INTERESTS & MENTORSHIP

I study how humans shape animal movement. My long-term goal is to learn what drives the movement of individual animals in response to human activity and disturbance, which is becoming a permanent and pervasive feature of our modern landscapes, including those with and around Calgary. Rather than simply documenting how human disturbance impacts animal movement and distributions, my research aims to determine how individuals integrate information about environmental pressures, resource opportunities, and their physiological states when deciding where to move. I use mostly non-invasive approaches and tools, including behavioural observations to study small-scale movement, camera traps to track movement over larger scales, physiological data extracted from fecal and hair samples to connect movement to physiological state.

I have mentored the following students:

Sophia Vig-Benko

RESEARCH ASSISTANT

Summer 2026

Luis Fuenmayor

RESEARCH ASSISTANT

Summer 2026

Jazmin Singh-Kaur

BIOL 5201 (PROJECT: TEMPORAL CHANGES IN RACCOON ACTIVITY)

Winter 2026

TEACHING EXPERIENCE & INTERESTS

I currently teach in the area of ecology and evolution. I am passionate about teaching essential quantitative skills to students, demonstrating through hands-on exercises that math and coding can also be fun. I have also developed courses in population and community ecology. While emphasizing theory, I also encourage my students to observe nature, which I believe is essential for generating good research questions. I have taught the following courses:

BIOL 4320: Field Biology Research Techniques

MOUNT ROYAL UNIVERSITY

2026

BIOL 3401: Big Questions & Big Data

MOUNT ROYAL UNIVERSITY

2026

BIOL 2213: Principles of Ecology & Evolution

MOUNT ROYAL UNIVERSITY

2025-26

BIOL 4330: Population & Community Ecology

MOUNT ROYAL UNIVERSITY

2025

BIOL 3350: Data Analysis in Ecology

UNIVERSITY OF MANITOBA

2023-2024

BIOL 1010: Biological Diversity & Interaction

UNIVERSITY OF MANITOBA

2023

Peer-reviewed publications

IN REVIEW/REVISION

1. Whaley, Z. B., Alston, M. A., Bahnson, C., **Newediuk, L.**, & Others. (2026). Elk without borders: Genomics reveals asymmetrical gene flow among a patchily-distributed cervid. *Ecological Applications*, in review.
2. Peers, M. J., **Newediuk, L.**, Boyle, S. P., JG, H., & Others. (2026). Seasonal timing and maternal condition influence the costs of reproduction in a cyclic species. *Journal of Mammalogy*, in revision.
3. Funk, K. R., **Newediuk, L.**, Brown, L. R., Ganswindt, A., & Vander Wal, E. (2026). Sociality shapes context-dependent stress responses to anthropogenic disturbance in african elephants. *Behavioral Ecology*, in revision.

PUBLISHED

1. **Newediuk, L.**, Jesmer, B. R., Mastromonaco, G., & Vander Wal, E. (2026). Landscapes of fearlessness revealed through hormonal responses to putatively risky places. *Behavioral Ecology*, 37, arag004. <https://doi.org/10.1093/beheco/arag004>
2. Oliver, R. Y., Yanco, S. W., Others, **Newediuk, L.**, & Others. (2026). Interacting effects of human presence and landscape modification on birds and mammals. *Science*, 392, 879–884. <https://doi.org/10.1126/science.adq3396>
3. Bath, D., & **Newediuk, L.** (2026). Revisiting size selective mortality in young fish: Do small teleosts really pay a cost? *Canadian Journal of Fisheries and Aquatic Sciences*, 83, 1–11. <https://doi.org/10.1139/cjfas-2025-0212>
4. **Newediuk, L.**, Richardson, E. S., Bohart, A. M., Roberto-Charron, A., & Others. (2025). Designing epigenetic clocks for wildlife research. *Molecular Ecology Resources*, 25, e14120. <https://doi.org/10.1111/1755-0998.14120>
5. Korpach, A. M., de Greef, E., **Newediuk, L.**, Schmidt, C., & Others. (2025). Moving to mate? Migration strategy does not predict genetic structure or diversity in bats (chiroptera). *Biological Journal of the Linnean Society*, 145, blae068. <https://doi.org/10.1093/biolinnean/blae068>
6. Boyle, S. P., Balsdon, M., **Newediuk, L.**, Litzgus, J. D., & Others. (2025). Anuran carcass persistence on roads: Causes and implications for conservation. *The Journal of Wildlife Management*, 89, e22731. <https://doi.org/10.1002/jwmg.22731>
7. **Newediuk, L.**, Mastromonaco, G. F., & Vander Wal, E. (2024). Associations between glucocorticoids and habitat selection reflect daily and seasonal energy requirements. *Movement Ecology*, 12. <https://doi.org/10.1186/s40462-024-00475-9>
8. **Newediuk, L.**, & Bath, D. R. (2023). Meta-analysis reveals between-population differences affect the link between glucocorticoids and population health. *Conservation Physiology*, 11, coad005. <https://doi.org/10.1093/conphys/coad005>
9. Collins, S. M., Hendrix, J. G., Others, **Newediuk, L.**, & Others. (2023). Bibliometric investigation of the integration of animal personality in conservation contexts. *Conservation Biology*, 37, e14021. <https://doi.org/10.1111/cobi.14021>
10. **Newediuk, L.**, & Vander Wal, E. (2022). Predicting the individual identity of non-invasive faecal and hair samples using biotelemetry clusters. *Mammalian Biology*, 102, 685–700. <https://doi.org/10.1007/s42991-021-00173-8>

11. **Newediuk, L.**, Prokopenko, C. M., & Vander Wal, E. (2022). Individual differences in habitat selection mediate landscape level predictions of a functional response. *Oecologia*, 198, 99–110. <https://doi.org/10.1007/s00442-021-05098-0>
12. Street, G. M., Potts, J. R., Others, **Newediuk, L.**, & Others. (2021). Solving the sample size problem for resource selection functions. *Methods in Ecology and Evolution*, 12, 2421–2431. <https://doi.org/10.1111/2041-210X.13701>
13. **Newediuk, L.**, Ethier, J. P., Boyle, S. P., Aubin, J. A., & Others. (2021). Sociopolitical factors drive conservation planning timelines: A canadian case study with global implications. *Biological Conservation*, 257, 109091. <https://doi.org/10.1016/j.biocon.2021.109091>
14. **Newediuk, L. J.**, & Hare, J. F. (2020). Burrowing richardson's ground squirrels affect plant seedling assemblages via environmental but not seed bank changes. *Current Zoology*, 66, 219–226. <https://doi.org/10.1093/cz/zoz047>
15. **Newediuk, L. J.**, Waters, I., & Hare, J. F. (2015). Aspen parkland pasture altered by richardson's ground squirrel (*urocitellus richardsonii sabine*) activity: The good, the bad, and the not so ugly? *Canadian Field Naturalist*, 129, 331–341. <https://doi.org/10.22621/cfn.v129i4.1755>

Conferences

INVITED TALKS

1. **Newediuk, L.**, Richardson, E. S., Biddlecombe, B. A., Kassim, H., Nicholas Lunn, L. K. ad, Rivkin, L. R., Salama, O. E., Schmidt, C., Jones, M. J., & Garroway, C. J. (2025). Climate change, age acceleration, and the erosion of fitness in polar bears. In *MRU's 2025 Sustainability Forum (Oral presentation)*. Mount Royal University.
2. **Newediuk, L.**, Schmidt, C., & Garroway, C. J. (2023). Managing landscapes for diversity and disease resilience. In *US Fish and Wildlife Service Pacific Regional Health Program Online Symposium (Oral presentation)*. Online.

PRESENTATIONS

1. **Newediuk, L.**, Melvin, Z., Thomas, L., & Wal, E. V. (2026). When blood is not thicker than water: Social bonds increase use of risky habitat more than kinship. In *Canadian Society for Ecology and Evolution Annual Meeting (Oral presentation)*. Toronto, Ontario.
2. Elizabeth Patterson, M. J., Levi **Newediuk**. (2026). Epigenomics as a tool for assessing well-being: A case study in zoo-housed polar bears. In *Canadian Society for Ecology and Evolution Annual Meeting (Oral presentation)*. Toronto, Ontario.
3. Whaley, Z., Bahnson, C., **Newediuk, L.**, LaSharr, K., Carstensen, M., & Seaborn, T. (2025). Elk without borders: Genomics reveals asymmetrical gene flow among patchily-distributed herds in manitoba, minnesota, and north dakota. In *The Wildlife Society Annual Conference (Oral presentation)*. Edmonton, Alberta.
4. Whaley, Z., Bahnson, C., **Newediuk, L.**, LaSharr, K., Carstensen, M., & Seaborn, T. (2025). Where the elk herd roams: Genomics reveals the status of connectivity between patchily distributed herds. In *International Association of Landscape Ecology - North America (Poster presentation)*. Raleigh, North Carolina.
5. Patterson, E., Petersen, S., Garroway, C., & **Newediuk, L.** (2025). An epigenomic approach to the long-term well-being assessment of polar bears in human care. In *Canadian Society for Ecology and Evolution Annual Meeting (Poster presentation)*. Sherbrooke, Quebec.
6. **Newediuk, L.**, Richardson, E. S., Biddlecombe, B. A., Kassim, H., Nicholas Lunn, L. K. ad, Rivkin, L. R., Salama, O. E., Schmidt, C., Jones, M. J., & Garroway, C. J. (2025). Climate change, age acceleration, and the erosion of fitness in polar bears. In *Canadian Society for Ecology and Evolution Annual Meeting (Oral presentation)*. Sherbrooke, Quebec.
7. **Newediuk, L.** (2025). Elk on the edge: A case study in wildlife research across organizational and jurisdictional boundaries. In *Manitoba Chapter of The Wildlife Society Annual General Meeting (Oral presentation)*. Winnipeg, Manitoba.
8. Bath, D., **Newediuk, L.**, Schmidt, C., & Garroway, C. (2025). Are there generalizable effects of habitat fragmentation on genetic diversity in vertebrates. In *Canadian Society for Ecology and Evolution Annual Meeting (Oral presentation)*. Sherbrooke, Quebec.
9. **Newediuk, L.**, Schmidt, C., & Garroway, C. J. (2023). Movement behaviour links genetic diversity to the environment. In *Canadian Society for Ecology and Evolution Annual Meeting (Oral presentation)*. Winnipeg, Manitoba.

10. **Newediuk, L.**, Mastromonaco, G. F., & Vander Wal, E. (2022). An adaptive role for glucocorticoids in habitat selection and specialization. In *Canadian Society for Ecology and Evolution & Ecological Society of America Annual Joint Meeting (Oral presentation)*. Montreal, Quebec.
11. **Newediuk, L.**, Mastromonaco, G. F., & Vander Wal, E. (2021). Glucocorticoid-mediated life history trade-offs via habitat selection. In *Canadian Society for Ecology and Evolution Annual Meeting (Oral presentation)*. Online.
12. **Newediuk, L.**, Prokopenko, C. M., & Vander Wal, E. (2020). Cross-scale functional responses improve inference in a resource selection function framework. In *Ecological Society of America Annual Meeting (Oral presentation)*. Online.
13. **Newediuk, L.**, Prokopenko, C. M., & Vander Wal, E. (2019). When individual differences matter for predicting population distributions. In *The Wildlife Society Annual Conference (Oral presentation)*. Reno, Nevada.
14. **Newediuk, L.**, Prokopenko, C. M., Avgar, T., & Vander Wal, E. (2019). Using movement to distinguish social and spatial influences on habitat selection behaviour. In *Canadian Society for Ecology and Evolution Annual Meeting (Oral presentation)*. Fredericton, New Brunswick.

SYMPOSIUM ORGANIZATION

1. Bowman, J., & **Newediuk, L.** (2026). Leveraging long-term research for one health. In *Canadian Society for Ecology and Evolution Annual Meeting (Symposium)*. UBC, British Columbia.
2. **Newediuk, L.**, Bowman, J., Boudreau, M., Garroway, C., Webber, Q., & Wishart, A. (2024). Supporting the next generation of long-term ecological research in Canada. In *Canadian Society for Ecology and Evolution Annual Meeting (Symposium)*. UBC, British Columbia.

Other research outputs

MEDIA COVERAGE

1. Diodati, H. (2026). The scoop on elk poop. *MRU News*. [link](#)
2. Dela Torre, C. (2026). Interview: Elk poop research. *CBC Calgary*. [link](#)
3. Wood, D. (2026). No. 22 priority: MRU researchers asking for help with elk study. *CTV Calgary*. [link](#)
4. Jarman, H. (2025). Study: Climate change causing polar bears to age faster. *MRU News*. [link](#)
5. Dolgin, E. (2025). The clock is ticking: How epigenetics could help save wildlife from collapse. *Knowable Magazine*. [link](#)
6. Zipp, K. (2025). Climate change is speeding up polar bear aging. *TWS Wildlife News*. [link](#)
7. Woodford, J. (2024). Polar bears appear to be ageing faster as the arctic gets warmer. *NewScientist*. [link](#)

Professional & academic service

COMMITTEE ROLES (LEADERSHIP)

Mount Royal University

BIOLOGY STUDENT RESEARCH COORDINATOR

2026-2029

Canadian Society for Ecology & Evolution Long-term Research Section

CHAIR

2026-2028

The Wildlife Society Newfoundland Chapter

PRESIDENT ELECT/PRESIDENT

2018-2020

COMMITTEE ROLES (PARTICIPATION)

Mount Royal University

ANIMAL CARE COMMITTEE

2025-2029

Mount Royal University

FST COMMITTEE ON INCLUSION, DIVERSITY, EQUITY, AND ACCESSIBILITY

2025-Present

Mount Royal University

BIOLOGY STANDING COMMITTEE FOR SESSIONAL APPOINTMENTS

2025-Present

OTHER SERVICE

- Second reader for two MRU Biology research students in 2025-2026 academic year

- Adjudicated posters for MRU Research Days 2026
- Reviewed 5 manuscripts for peer-reviewed journals in 2026 (24 since 2020)

Grants & research funding

EXTERNAL FUNDING

\$23,900 (Mount Royal University)

ALBERTA CONSERVATION ASSOCIATION RESEARCH GRANTS

2026 (one year)

\$81,400 (Memorial University of Newfoundland)

MANITOBA HABITAT HERITAGE CORPORATION FISH & WILDLIFE ENHANCEMENT FUND

2018 (over two years)

\$15,000 (Memorial University of Newfoundland)

NATURE CONSERVANCY OF CANADA SCIENCE SUPPORT PROGRAM

2018 (over two years)